

LOCAL ADDENDUM TO FORMULA SAE 2025 RULES V3.0 – May 23, 2025

This Addendum defines Revisions and Clarifications versus the International 2025 Formula SAE-I Rules for the December 2025 Formula SAE-A competition.

A Summary of the Changed Clauses in the FSAE-I Rules for 2025 versus 2024 has been circulated to teams and posted to the local Formula website. These may not identify all of the detail changes so teams must ensure they are familiar with FSAE-I rules for 2025 as well as this Addendum.

Except where otherwise noted, this Addendum applies to both Internal Combustion Engine Powered Vehicles and Electric Vehicles.

Any new items or significant changes in this 2025 Addendum from the final version of the prior year 2024 Local Addendum, are highlighted in blue text.

Teams should particularly note the local requirements relating to On-site Registration, Technical Inspection and Driver Requirements in the Applicable sections of this Addendum.

Note: Wherever it is not clear as to application of the rules to a proposed vehicle design, or a radically new concept is proposed for a vehicle, the team should submit the concept to the Rules Committee in advance and not rely only on the team interpretation of the rules, as they may not have been drafted with this proposed new concept in mind.

Please read all the rules carefully. The Change Log on Page 26 will identify subsequent revisions.

OVERVIEW

Scoring

At the Australian event, the general approach will be that Electric and IC Vehicles will compete in the same events, with scores then applied separately to the IC and Electric vehicle classes to determine placings under separate IC and EV Classes. Note that dependent on the number of entries, additional place scoring and prizes beyond the 1st place, may not be available for IC Vehicles. Further details of the awards and scoring will be included in the Event Handbook after Registrations close.

The Maximum Score, and Your Score, and calculation method for will be in line with the [2025 FSAE international Rules](#) other than for the Cost and Design Events.

Vehicle Eligibility

Any team wishing to enter a 2nd Year Car for 2025 must follow the requirements for the Australasian Competition as defined in the Clauses under GR.9.3.2 in this Addendum and must submit the application by the specified date prior to the close of Registration.

Document Submission:

Required documents are listed in Table DR-1, Clause EV.2.3 and the Appendix PDA - 1.

All electronic document submissions must be uploaded by the Team Leader using the email address that was used to register your team.

If you have multiple team entries, please upload your documents for each entry separately using a unique email address.

All documents must be uploaded to <https://forms.gle/mjrx1NDydSmAhByV8>

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Submissions will only be accessible by SAE-A event organisers, judges and nominated persons.

ELECTRIC VEHICLES

Voltage Definitions

It should be noted that the definitions used in the Formula SAE Rules for High and Low Voltage do not necessarily correlate with the Definitions used by Electrical Authorities in different countries or jurisdictions. In Formula SAE, High Voltage is used to indicate voltage levels which if not adequately isolated or insulated may cause injury or fatality if contacted without wearing adequate protection suitable for the voltage level of the vehicle High Voltage systems

State of Charge

In line with best practice safe handling requirements for Accumulator / Battery packs, it is recommended that Electric Vehicle Accumulators are transported to the competition site with a SoC of not more than 50%.

All teams must be able to accurately identify accumulator state of charge during technical inspection and must also present for approval at Technical Inspection their method of safely charging and discharging their battery pack. The system should be capable of fully discharging the battery pack in less than 8 hours.

Electric Vehicle Accumulators may not be charged on-site until passed by EV technical inspection or as directed by the EV technical inspectors and must be on-site at the time of team Registration.

Parc Fermé

Following each vehicle's completion of the Endurance event, the vehicle will be impounded in a "Parc Fermé" where further inspection may be carried out on the vehicles so impounded. This will also apply to all IC and EV cars even if classified as DNF.

Any temperature monitoring device and the Energy Meter will be removed for EV cars in Parc Fermé.

No team members will be allowed to access their vehicle while it is impounded or located in "Parc Fermé", except under the direction of the officials. See further under Dynamic Events - Endurance.

Rules Enquiries

If you have any questions regarding interpretation of the competition rules, please follow steps 1 and 2 below before submitting a question to the Rules Committee.

1. Check the [Formula SAE-A Rules Q&A document](#) to see if your query has already been answered.
2. Ask your Team Leader or Faculty Advisor. In many cases they will be able to assist you.
3. Submit a question to the Rules Committee using the ["Ask a Question" link](#) on the SAE-A website.

The Rules Committee may require further information from the team prior to finalising an answer and may also require review of the team's FMEA for major variations or new concepts.

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Resources

Document templates, forms and guidelines are available from the [SAE US](#) and [SAE Australasia](#) websites. The Local Website will include all unique locally required forms and additional information documents. Teams should also refer to and use all available resources on both the US and local websites in order to establish compliant and effective designs.

GR-GENERAL

US RULE	PAGE	CHANGES & CLARIFICATIONS
GR.2.4	6	Restrictions on Vehicle Use Add: The following further clarification to the US Rules should be noted: These vehicles are not assumed to be capable of performing in other environments, nor other types of competition, where the speed and physical limitations of the Formula SAE competitions evaluation courses, are removed.
GR.3	6	Team Responsibility Add Additional Clause for Australasian Event.
GR.3.6	7	Risk Assessment and Safe Working
GR.3.6.1	7	All teams are expected to operate under the OH&S, and any other applicable safety practices or requirements of their respective university. Teams must prepare a Risk Assessment and Hazard Reduction Summary relating to working on their vehicle, including potential hazards in the components installed on their vehicle. Teams should also ensure that their Risk Assessment and reduction response are applied beyond their university site in any work done off-site from the university and in the pits or to their vehicles at the competition event.
GR.3.6.2	7	All teams must submit a soft copy of their RA & HR Summary and safe working practices in line with Clause GR.3.6.1, which should be in accordance with their university policies and procedures. The Summary will vary by team, according to their university's requirements and applicable regulatory authorities. There is no mandatory pro-forma for this document. Existing requirements or procedures that teams are operating under can form the basis of this summary. The date for submission of this Summary is shown in the Appendix PDA-1.
GR.3.6.	7	Electric Vehicle Teams. In addition to the above Summary at on-site registration, teams shall provide a statement signed either by their Faculty Advisor, or an alternative appropriate University representative, that confirms their adherence to the requirements of Sections 1, 2 and 3 as detailed in the 2024/25 Event Handbook and that their vehicle, equipment and operating procedures meet or exceed accepted professional standards for Electric Vehicle safety. EV entries will not be allowed to proceed to technical scrutineering until this document has been received by an SAE-A event official., The form for this is provided by SAE-A on the website and can be downloaded from the following link: Electric Vehicle Safety Form

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GR.4		RULES AUTHORITY
GR.4.6	7	Add Clause a) The SAE-A event will be held under the International Sporting Code of the FIA, the National Competition Rules of Motorsport Australia (formerly CAMS), and the Speed Event Standing Regulations, any relevant Championship Sporting Regulations as approved by Motorsport Australia (MA), these Supplementary Regulations and any Further Regulations and instructions to competitors that may be issued. b) The event shall be a Formula SAE Inc. Club Meeting run under the current year F-SAE Rules and F-SAE-A Rules Addendum. c) This Event will be conducted in compliance with Motorsport Australia OH&S and Risk Management Policies, which can be found on their website at https://motorsport.org.au
GR.8.5	9	Protest Period Delete US words and Add: Protests concerning any aspect of the competition must be filed within 60 minutes of the posting of the scores for the event to which the protest relates unless an alternative period is announced by the officials. For Endurance/Efficiency and Overall results, the protest review may be heard the following day, but the protest must be lodged within the nominated 60 minutes on the Sunday
GR.8.6	10	Decision Add following words to clause: The information that is acceptable to be considered by the judges in reaching a final decision is entirely at the discretion of the judges. Event Officials will be “Judges of Fact” in relation to any protest. Any material that is not supported by FSAE-A’s official recording of data or reports will be excluded.
GR.9.3.2	10	FSAE Competition Eligibility 2nd Year Vehicles Delete the US words and Add the following for the FSAE-A Competition: Vehicles that have competed during any one (1) previous Formula SAE year may compete provided that they have been substantially modified from their first competition. Teams proposing to enter a 2nd Year Vehicle must submit a request a minimum of six weeks in advance of the Entry Registration final date in order to be considered for acceptance by the Competition Organisers. Refer Appendix PDA-1 for Dates. This should include a general overview of the reasons behind the proposal and summary of intended changes, but full finalised design details are not required at this stage. Use the form “Proposal to compete with 2nd Year Car” to be available on the FSAE-A website. Any non-compliance with the current year Rules must be defined by the team and submitted as a Rules Enquiry for review and acceptance. All Safety requirements must be complied with.

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GR.9.3.2 (Cont'd.)	10	If accepted, subsequent photographic and design documentation detailing the final modifications are required along with a statement from the team's Faculty Advisor to be submitted. Refer Appendix PDA-1. Penalties for insufficient redesign or insufficient knowledge by the team may be applied during the Design Event. Refer to the Rule S.4.11 "Potential Penalties for Insufficient Redesign".
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AD-ADMINISTRATIVE

US RULE	PAGE	CHANGES & CLARIFICATIONS
AD.3.1.4	12	The US Rules State: Each team member may participate at a competition for only one team. This includes competitions where the University enters both IC and EV teams. For clarification, the intent of this rule is that the members competing/presenting in each of the Static and Dynamic events at the competition must be designated as either part of the EV or IC team and cannot cover both vehicles. This does not exclude team members from working on both vehicles or providing other support beyond the designated events.
AD.3.2	12	Age Delete US Rule and Add Any defined University Student Team member below the age of 18 years must present a signed declaration by their parent or guardian. A copy of the Age Declaration form will be available on the FSAE-A website.
AD.3.3	12	Driver's Licence and Competition Licence Add: All Drivers of each team must hold the minimum of a Motorsport Australia SPEED Licence; see NCR 47 or the equivalent authority issued by Motorsport Australia. International drivers must apply for an MA SPEED licence and obtain an 'Authority to Compete' from MA. The issue of MA licences will be coordinated by SAE-A for all new licences, via team members filling out the Registration Form. Drivers with existing Motorsport Australia licences should renew their licence directly with Motorsport Australia.
AD.3.4	12	Society Membership Delete US words and Add: Formula SAE-A is open to teams from Australia/NZ universities, TAFE colleges and some overseas teams. All members of Australia/NZ teams must be members of SAE-A. Team members of international teams must be members of their local SAE Society, ATA, IMechE or VDI. If no local society membership is available, they must apply to become members of SAE-A in order to compete at the event. Students can apply to join SAE-A online at: www.saea.com.au. Proof of membership, such as a receipt for membership payment or current membership card, can be provided via electronic upload at the time of registration, or must be presented at the competition.

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AD.3.5	12	Medical Insurance Add: Individual medical insurance coverage per the US rule is obviously desirable but government versus private coverage varies significantly around the world. Accordingly, foreign and local teams must ensure that they are adequately covered by their domestic insurance and carry adequate travel medical and accident insurance to cover their time in Australia and at the competition.
AD.3.6	12	AD.3.6 Disabled Accessibility Delete USA words and Add Teams with members who may have special accessibility requirements, must contact the Event Organisers prior to the submission of the final team member list.
AD.4	12	Revise Heading to read: INDIVIDUAL & TEAM ON-SITE REGISTRATION REQUIREMENTS
AD.4.1	12	Delete US Clause
AD.4.2	12	On Site Registration
AD.4.2.3	12	Delete US words and add On Site registration should be completed as soon as possible after the team arrives on site.
AD.4.3	12	Added Clause: Team Member Registration - Documentation Required All participating team members must provide either at the time of online registration, or at the event on-site registration, the following documentation: • Photographic Identification: e.g. Government issued driver's licence, Government issued proof of age card; passport; University ID card. All forms of photographic identification must be valid at the time of submission. • Emergency Contact Information: Each team member must provide the name and phone number of a designated emergency contact. • Proof of Society Membership, as per AD3.4 above. • For Team Drivers, Proof of a valid CAMS licence or equivalent authority, as per AD3.3 above. • Hazardous Materials Binder with MSDS sheets.
AD4.4	12	Added Clause: When teams arrive at the FSAE-A venue and register, both the Team Captain and the Faculty Advisor (and OH&S representative, if not the Faculty Advisor) must be present and be able to identify themselves as being those nominated in those roles at the initial online registration. At the on-site registration, all teams must submit a completed copy of the Technical Inspection Checklist as primary self-evaluation by the Team and a copy of the Egress Times List with the names of all drivers and the times they achieved in the Egress test. The drivers required to complete the test for verification at TI will be advised at the event. This may or may not include all drivers.

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ADR.4.4 (Cont'd.)	12	All EV teams must also submit a completed copy of the Electrical Inspection Checklist. This must be signed by both the Team Captain and the Faculty Advisor. NOTE: • To avoid late issues all teams should carry out an internal initial compliance evaluation against the TI Sheets 6 weeks prior to the event. • The team must present the final completed Checklists and Egress List at Technical / Electrical Inspection. Teams should complete on-site registration by the designated time. If not completed by 9:00 AM on the Thursday of the Competition at the latest, a penalty of 40 points, will be deducted from their overall score.
AD.5.1 AD.5.1.4 AD.5.1.5	12 13 13	Faculty Advisor Add the following two clauses: To improve communication and avoid duplication, all contact between teams and SAE-A officials prior to the event should be in accordance with the <u>FSAE-A Team Communications Protocol</u>. The FA can often help with prior knowledge and interpretations and ensure maximum efficiency in contacts; they may also liaise with other Faculty Advisors. All teams must have a designated OH&S Advisor responsible to ensure compliance with each university's OH&S practices and to ensure the FSAE-A event requirements are also met. The Faculty Advisor shall be the designated OH&S Advisor unless another person is designated by the university to fill this role and SAE-A is notified in writing of their appointment at least two weeks prior to the event. The designated person must attend all days of the event.
AD.5.2.1	13	Add the following to the US words. If more than one ESO is trained/appointed, a Principal ESO must be designated who shall be responsible for signing any documentation, any liaison with officials, and whose details will be shown where required under the rules.
AD.6 AD.6.2.1 AD.6.2.2 AD.6.2.3 AD.6.2.4	14	COMPETITION REGISTRATION General Information. Registration Details. Registration Fees Waitlist Delete US words for these 4 clauses and add: Formula SAE-A is open to teams from Australia/NZ Universities and TAFE Colleges and some overseas teams. Registration is via the online registration link only. If more than 36 applications are received (32 Max. AV/EV) there may be a limit imposed. This will be monitored and determined by 31 July 2025. If the number of entries exceeds the maximum available event number, then a ballot or other method will be used to reduce the number of overseas entries within the available number of entrants. If a reduction is required to the number of entries, this decision will be announced to the affected overseas teams as soon as possible after the entry closure date.

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AD.6.5	14	Withdrawals Delete US words and add: Any team registered for the Australasian competition must notify the organisers via formulasae@sae-a.com.au as soon as any decision is made to withdraw in order to allow other teams the opportunity to compete. Any team which has submitted an EOI for potential entry, but not yet registered must advise via formulasae@sae-a.com.au that they will not be registering as soon as such decision is reached.
AD.7	14	COMPETITION SITE
AD.7.5	15	Add clauses:
AD7.5.1	15	Fuels, Fluids and Energy Storage:
		Internal Combustion engine vehicles must be drained of fuel before entering the event site for safety and also as only event supplied fuel is to be used.
AD.7.5.2	15	Electric Vehicle Accumulators should preferably be discharged to 50% or less state of charge before entering the event site and may not be charged until passed by EV technical inspection or as directed by the EV technical inspectors.
AD7.5.3	15	Draining of Fluids. No fluids are to be drained within the pit area except into approved receptacles and no fuels/oil are to be drained in the pit area without prior approval from the organisers and with appropriate fire protection present. Fluid Containers. No open vehicle fluid containers are allowed in the pit area. No fuel or other flammable liquids to be stored on site.

DR-DOCUMENT REQUIREMENTS

US RULE	PAGE	CHANGES & CLARIFICATIONS
DR.2.1	16	Submission Location
DR.2.1	16	Delete USA Clause and Add: Teams entering Formula SAE-A competitions in Australia must upload the required documents to the designated submission site as defined in Appendix PDA-1 of this Addendum.
DR.2.2	17	Submission Details.
DR.2.2.5	17	Add new clause: Submissions must adhere to standard naming Car No_ University Name_ses.xls/IAD.pdf/spec.XLS/Design.pdf/Design.mp4/Cost_Report.xls/Cost_Report.pdf/etc. and file format to be uploaded to https://forms.gle/mjrx1NDydSmAhByV8
Tables DR	18	Delete US tables and add the new DR-1 and DR-2 Tables as follow:

Table DR-1 Submission Information

Use the template file or form available on the SAE-A website **AD.2.2.1**

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Submission:	Refer to:	File Format:	Group:
Structural Equivalency Spreadsheet(s) (SES) as applicable to your design including the integrated AID report.	F.2.1	XLSX	Tech
ETC – Notice of Intent	IC.4.3	PDF	ETC
ETC- Systems Form (ETCSF)	IC.4.3	XLSX	ETC
EV – Electrical System Advisor and Electrical System Officer Form	AD.5.2 AD.5.3	PDF	Tech
EV - Electrical System Form (ESF)	EV.2.1	XLSX	Tech
Cost Report	PDA-2 S.3.4.2	PDF	Cost
Cost Report Support Documentation	PDA-2 S.3.4.2	XLSX	Cost
Cost Amendment Report	PDA-2 S.3.7.2	PDF	None
Design Briefing	S.4.3	PDF	Design
Vehicle Drawings	S.4.4	PDF	Design
Design Specification Sheet	S.4.5	XLSX	Design

Table DR-2 Submission Penalty Information

Penalty Group	Penalty Points Per Day	Maximum Point Penalty	Not Submitted within X days of deadline
Tech	- 5	- 50	Removal of team from applicable event or may be removed from the total competition. X = 21 Calendar Days
ETC	Not approved to use ETC. See DR.3.4.1		
Cost	- 5	- 80	Team will receive 0 points for the Cost Report. Refer S.3.6.1 X = 14 Calendar Days
Design	-5	-50	Removed from Design Event. Score 0 points in Design Event X = 21 Calendar Days

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US RULE	PAGE	CHANGES & CLARIFICATIONS
DR	16	Documentation Requirements: Additional Clauses for Australasian Event:
DR.4	18	Hazardous Materials
DR.4.1	18	In addition to the documents defined for the US event, FSAE-A requires submission of Hazardous Materials Statements and related Material Safety Data Sheets (MSDS) in line with local OH&S practices. The MSDS lists the hazardous ingredients of a product, its physical and chemical characteristics (e.g. flammability, explosive properties), its effect on human health, the chemicals with which it can adversely react, handling precautions, the types of measures that can be used to control exposure, emergency and first aid response. For more information refer to local State Authorities or the Australian Government Safework Australia website.
DR.4.2	18	The team must prepare a binder listing all hazardous materials and attach the MSDS for all the nominated materials. A preliminary listing should be submitted by the nominated date in Appendix PDA-1 with a final version brought to Team Registration at the Event.

V-VEHICLE REQUIREMENTS

US RULE	PAGE	CHANGES & CLARIFICATIONS														
V.3.1.1	20	Suspension. Clarification of intent: Delete US words and Add: The vehicle must be equipped with fully operational front and rear suspension systems including shock absorbers and a usable wheel travel of at least 50mm, and a minimum jounce of 25mm, with driver seated.														
V.4.3.3	22	Tyre Set (Also relates to IN.14.2 on Changes allowed post Inspection)														
V.4.3.3.c	22	Add additional clarification clause Tyres and Compounds: 1. Teams should have two sets of tyres – a Dry set and a Wet set. 2. Within each vehicle set, the tyres do not have to be identical composition or size but must be identical Right to Left at Front and Rear but Front and Rear can differ. 3. Teams can have extra Dry or Wet tyres but they must not introduce any difference to the basic sets of tyres as approved at Technical Inspection; a new compound or size cannot be introduced on any replacement tyre that is put onto the vehicle. The replacement tyre must be identical to the tyre it is replacing on the vehicle at that particular corner of the vehicle. 4. If there is a difference in tyres within a TI approved set of Dry (or Wet) tyres, then if replaced each tyre must maintain that same combination as presented at TI. 5. So if the following exists at TI, <table><tr><td>Vehicle Position</td><td>Front LH</td><td>Front RH</td><td>Rear LH</td><td>Rear RH</td></tr><tr><td>Tyre Type</td><td>A</td><td>A</td><td>B</td><td>B</td></tr></table> then any front tyre must be replaced with an A and a rear tyre with B; an A tyre cannot replace a B tyre or vice-versa.					Vehicle Position	Front LH	Front RH	Rear LH	Rear RH	Tyre Type	A	A	B	B
Vehicle Position	Front LH	Front RH	Rear LH	Rear RH												
Tyre Type	A	A	B	B												

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F-CHASSIS AND STRUCTURAL

US RULE	PAGE	CHANGES & CLARIFICATIONS
F.5.6.3	32	Added Clarification Any bends in the front view, may be supported by transverse braces as an alternative to triangulation into the Primary Structure. Teams should obviously try to avoid any front view bends, other than as required at the top of the hoop and at the upper attachment to the upper Side impact Structure/monocoque.
F.7.8	41	Monocoque Attachments - Explanatory Note: When completing the SES where it requires the “Distance to nearest edge” Cell to be completed, the following is the definition for the dimension to be used and which teams should use in any of their calculations. “Distance to nearest edge” is the dimension from the centre line of the attachment bolt to the nearest monocoque free edge. This dimension must be taken from whichever of the bolt holes is closest to the nearest free edge.
F.7.8.2	42	Note that the clause 7.8.2 does not negate any of the requirements for the Main Hoop attachments in 7.8.1 and is intended to cover a monocoque which has an open rear or where the monocoque is continuous across the rear joining/closing the two sides.
F.7.8.5/7	42	Inserts for Critical Fasteners Note: There is an error in the 2025 Monocoque/Hybrid Non-Ferrous SES under the ‘F.7 Composite Chassis’ Tab. In the Rear Bulkhead section on Rows 164 and 165 it incorrectly calls for ≥ 12 GPa and 2 GPa for the Insert Compressive Strength and Shear. This should read 12 MPa and 2 MPa.
F.8.2	43	Anti Intrusion Plate – AIP - Explanatory Notes When completing the SES for an AIP attached via bolting to a monocoque bulkhead the inputs are to ensure an adequate thickness of backing plate supporting the bulkhead to ensure the 15 kN tear out loads are achieved. In the input lines Backing plate thickness and Backing plate perimeter on bulkhead are referring to a backing plate or load spreading washer supporting the bulkhead inserts, not to the AIP dimensions. Minimum Fastener spacing, edge or corner distance must have the minimum distance from the centreline of the bolt to the nearest edge or a corner of the bulkhead, or the spacing between the mounting bolts (at 50 mm Min per F.3.1.b)
F.8.2.3	45	Attachment of the Anti Intrusion Plate For clarification of the SES. While not listed in the Rules, the SES allows for use of a Quick Release mechanism for attaching the AIP to the Front Bulkhead. A Quick Release mechanism is not allowed for attaching the Impact Attenuator to the AIP. Any Quick Release mechanism must ensure it cannot be accidentally operated and must ensure that the minimum required load of 120 KN is maintained in all directions except as noted for a slot design. Any shear pins must extend through the thickness of the AIP. If a slotted design is used, the slots in the AIP must be at least a minimum length of 3 times the width to ensure adequate engagement. A minimum load of 95 KN must be achieved in the direction of removal from the slot.

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F.8.5	45	Impact Attenuator Attachment Add Note: Bonding is the preferred method of attachment of the Impact Attenuator to the Anti Intrusion Plate.
F.10.3	50	Cells and Segments; Refer to Guidelines posted to web site for clarification on intent of the rule change and completion of the SES input.

T – TECHNICAL

US RULE	PAGE	CHANGES & CLARIFICATIONS
T.1.5 -1.8 T.1.5.2	55-57 55	Driver's Seat - Add Additional Clause Seat Structure In addition to the requirements relating to seating and driver protection in clauses T1.5, T.1.6, T.1.7 and T.1.8, while no structural requirements are included in the Rules, GR.1.5 indicates that an adequately located and fixed seat structure is required to support the driver through a variety of loads during the event. For example, Seats/Drivers should not load the firewall unless the thickness and mounting of the firewall has taken the potential loads into consideration. Potential loads should be considered in the size and placement of related fasteners/fixings/brackets
T.1.6 T.1.6.4	56 56	Thermal Protection Add Clause In addition to when seated in normal driving position, the heat protection requirements also apply to areas where contact may be made on entry to, or exit from, the cockpit.
T.1.8	57	Tractive System Firewalls (EV Only) As the US 2024 rules no longer provide guidance as to the thickness of the AL Firewall, the following clarifies the expectations for the FSAE-A event. T.1.8 must be met with the firewall made from a rigid non-flammable material and rigidly mounted. To comply, the firewall itself should meet these requirements, without support from other materials and of adequate thickness. The Al must be a minimum of 0.5mm thickness and the traditionally used and readily available 1 mm thick Al sheet (or thicker) is preferred. Rigidity may be obtained by structural shaping and/or via the edge mounting or addition of Al bracing. The firewall should not deflect into the accumulator clearance zone under a reasonable load such as from a seated driver.
T.2.2.1	58	Harness Specification Add additional note to Clause d. FIA Specification 8853/98 While FSAE International Rules have deleted the FIA 8853/98 homologation. These continue to be permitted by Motorsport Australia, therefore, harnesses to this level will still be accepted at the Australasian event provided they are still

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T.2.2.1 (cont'd.)	58	within their validity date and have the FIA Hologram on all harnesses manufactured after 01/01/2013. Note that under both 8853/98 and 8853/2016 the shoulder straps must be three inches width if used without a HANS/FHR device, while two inch shoulder belts are acceptable only with a HANS/FHR. A HANS/FHR may be run with a 2 inch or 3 inch belt. Two inch belts are acceptable for the other straps (waist; crotch).
T.3.4	64	Brake Light Add sub-clause T3.4.5
T.3.4.5	64	To assist safety / fair play in the endurance event, any vehicle with a brake light illuminated continuously, or under non-braking conditions, will be black flagged.

VE-VEHICLE AND DRIVER EQUIPMENT

US RULE	PAGE	CHANGES & CLARIFICATIONS
VE.1.1	75	Vehicle Number Delete US words and Add: The assigned vehicle numbers must appear on the vehicle as follows: a) Locations: In three (3) locations: the front and both sides b) Height: 150 mm (6 inch) high c) Font: Helvetica Bold d) The numbers do not need to meet the US minimum separation of 18mm but they must not actually overlap. They must be easily differentiated from one another and readable from a distance of at least 100 metres. e) Colour: Day Glo Yellow on a black background f) Background shape: The number background must be one of the following: round, oval, square or rectangular. g) There must be at least 25.4 mm (1 inch) between the edge of the numbers and the edge of the background. h) Clear: The numbers must not be obscured by parts of the car, including, but not restricted to wheels, side pods and exhaust system.
VE.1.3	75	Logos: Delete US words and Add the following clause: The logos of the major sponsors of the competition, as well as the SAE-A logo, must be displayed on the upper surface nose cone of the vehicle, symmetric about the vehicle centreline, in a clear space of 210mm wide by 500mm long. The logo sticker applique will be supplied by the Event organisers SAE-A to teams at Registration at the Event. A typical example of the mounted Sponsor Logo applique is shown on the SAE-A website at https://www.saea.com.au/rules-documents-templates Failure to use and/or correctly mount the supplied logos will incur a minimum penalty of 10 Points.

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VE.1.4	75	Inspection Stickers A more compact Inspection Sticker will be used at the Australasian Event. Accordingly, Revise first bullet point of US words to a clear and unobstructed area, minimum 19 cm wide x 14 cm high																						
VE.1.5	75	Transponder Delete US words and Add: Transponders will be used for timing at the Formula SAE-A Event. These will be supplied at the event by the organisers and installed at the officials’ direction. They must be mounted to ensure a clear path for the signal between the transponder and ground.																						
VE.3	77	DRIVER EQUIPMENT Delete US Clauses VE.3.2.2; VE.3.3.1 to VE.3.3.6. The US Clause VE.3.3.7 (The requirement to wear Arm Restraints) is retained. VE.3.3.8 Add additional clause VE.3.3.8: Driver’s equipment must be worn that is in accordance with the following Schedule VE-3. This ensures optimum protection for drivers at the Formula SAE-Australasia event and teams need only refer to the MA Regulations, General Requirements, Schedule D for relevant details. The schedule below, VE-3, also ensures compliance (or above) with the Formula SAE US Rules. The Standard relevant to the Apparel item (Level A, B or C) is defined in Schedule D, Apparel, on the MA website. Go to Schedule D at https://motorsport.org.au/wpblog0fe832abcb/wp-content/uploads/docs/default-source/manual/general-requirements/2024/2024-schedule-d.pdf to download the latest PDF update and refer to Section 2, Apparel Standards, and Section 3. Frontal Head Restraints are not required for Formula SAE but are optional. If used, they must comply with the Motorsport Australia schedule. Schedule VE-3 The Minimum Requirements are: <table><tr><td><u>Apparel Item</u></td><td><u>Level</u></td></tr><tr><td>Helmets:</td><td>Level B</td></tr><tr><td>Frontal Head Restraint:</td><td>Level B</td></tr><tr><td>Overalls:</td><td>Level C</td></tr><tr><td>Underwear:</td><td>Level B</td></tr><tr><td>Balaclava:</td><td>Level A</td></tr><tr><td>Shoes:</td><td>Level B</td></tr><tr><td>Socks:</td><td>Level A</td></tr><tr><td>Gloves:</td><td>Level A</td></tr><tr><td>Plus :</td><td></td></tr><tr><td>Arm Restraints</td><td>SFI Spec 3.3 (minimum)</td></tr></table>	<u>Apparel Item</u>	<u>Level</u>	Helmets:	Level B	Frontal Head Restraint:	Level B	Overalls:	Level C	Underwear:	Level B	Balaclava:	Level A	Shoes:	Level B	Socks:	Level A	Gloves:	Level A	Plus :		Arm Restraints	SFI Spec 3.3 (minimum)
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IC- INTERNAL COMBUSTION ENGINE VEHICLES

US RULE	PAGE	CHANGES & CLARIFICATIONS
IC.5.1	834	Fuel
IC.5.1.4	84	Add additional clarification clause : - At the Australasian Event, the fuels supplied will be unleaded petrol with a Research Octane Number (RON) of 98 and Ethanol E85. - Teams must nominate the type of fuel required when they submit their entry registration. Notes: 1. E85 formulation and characteristics may vary between locations and between the fuel obtained by teams during development and that supplied at the event. All US rules relating to ethanol (restrictors, etc.) will apply. 2. All fuel must be drained from the vehicle prior to entering the competition site.
IC.5.4	84	Fuel tank Filler Neck and Sight Tube
IC.5.4.9	85	Add additional clause: The filler neck and sight tube must meet the positive fixing/retention requirements for fuel lines as per IC5.8.

EV-ELECTRIC VEHICLES

US RULE	PAGE	CHANGES & CLARIFICATIONS
EV.2	90	DOCUMENTATION
EV.2.3		Add new Clause FMEA Teams should prepare an FMEA as part of their vehicle design process for review by their ESA and team leader. The FMEA will not require to be formally submitted. Any FMEA that is prepared should be brought to the Design Event by the team.
EV.3.2 EV.3.2.1	90	Energy Meter Revise second sentence to read: Refer to the FSAE-A web site for detail information on the Energy Meter.
EV.3.2.4	90	Delete US Clause and Add following to clarify the event supplied meter installation. The Energy meter must be connected to the TSMPs on the TSMP side of the body protection resistors. Energy Meters will be retrieved from teams at parc fermé after the Endurance event (or earlier for teams that do not complete in the Endurance Event) and analysed after this retrieval.
EV.3.2.6	90	Add additional clause to clarify HV DC measurement cable colour. The energy meter HV DC measurement cable is exempted from the requirements of EV.6.3.4, to be of Orange colour. No additional covering beyond the existing white insulation sheath is required and should not be added.

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EV.4.3	92	Accumulator Containers
EV.4.3.9	92	Add Clause All items within the Accumulator Container including cables and wiring must be appropriately secured to prevent movement, rubbing or chafing.
EV.4.9	93	Housings and Enclosures
EV.4.9.3	93	Add Clauses If the Accumulator Container is constructed from a conductive material, the entire interior surface must be lined with a non-conductive material per EV.6.2 and EV.6.5.5
EV.4.9.4	93	Any conductive housings shields, structures or similar within in the Accumulator Container must have a low resistance connection to GLV System Ground
EV.4.10	923	Accumulator Hand Cart
EV.4.10.5	93	Add additional Clause The Hand Cart must be prominently labelled. The label on the Hand Cart must include the vehicle number; the university name; and the ESO phone number(s). The label must be written in Roman Sans-Serif characters of at least 10mm high on the lid or top of each Hand Cart. The characters must be clearly visible and placed on a high-contrast background.
EV.5.6	96	Precharge and Discharge Circuits
EV.5.6.6	96	Add Clause defining the PDOC PDOC. The components within the pre-charge and discharge circuits that dissipate heat (power resistors, linear MOSFETs, heatsinks etc.) must be monitored for thermal overload by a Pre-charge/Discharge Overload Circuit. In the case of a thermal overload, the PDOC must open the shutdown circuit before the components exceed their manufacturer’s recommended maximum operating temperature. This must be done without the influence of any programmable logic. See also EV.8 Shutdown Circuit regarding shutdown and reactivation of the tractive system after a fault. The status of the PDOC must be shown to the driver by a red indicator light in the cockpit that is easily visible even in bright sunlight. This indicator must light up, if the PDOC detects a thermal overload of the pre-charge or discharge circuit. The indicator light must be clearly labelled with “PDOC”. The PDOC may be omitted if the pre-charge and discharge circuit is designed for continuous operation in a faulted state and will not adversely affect nearby devices. If the PDOC is not fitted, theoretical and experimental evidence must be submitted to demonstrate that the pre-charge and discharge circuit cannot overheat to the point of damage to the vehicle and that the heat generated can be appropriately dissipated when fitted to the vehicle. Any failure modes must be documented in the FMEA with appropriate controls in place as required. If the PDOC is not fitted, then the rationale/evidence must be submitted concurrent with the FMEA timing.

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EV.5.10 EV.5.11 EV.5.12 EV.5.12.1	97	Tractive System Active Light – TSAL Delete the US Clauses EV.5.10/11 and: Add the following F.12 clauses The vehicle must include a Tractive Systems Active Light (TSAL) that must: a. Illuminate when the GLV System is energized to indicate the status of the Tractive System b. Be directly controlled by the voltage present in the Tractive System using hard wired electronics. Software control is not permitted. c. Not perform any other functions. The TSAL may be composed of multiple lights inside a single housing When the voltage outside the Accumulator Container(s) exceeds T.9.1.1 , the TSAL must: a. Be Colour: Red b. Flash with a frequency between 2 Hz and 5 Hz When the voltage outside the Accumulator Container(s) is below T.9.1.1 , the TSAL must: a. Be Colour: Green b. Stay continuously illuminated The TSAL mounting location must: a. Be near the Main Hoop at the highest point of the vehicle. b. Be inside the Rollover Protection Envelope F.1.13 c. Be no lower than 150 mm from the highest point of the Main Hoop. d. Not allow contact with the driver's helmet in any circumstances. e. Not be in proximity to any other lights. The TSAL must be visible: a. From every horizontal direction, except small angles which are blocked by the Main Hoop b. From a point 1.6 m vertically from ground level, inside a 3 m horizontal radius from the TSAL c. In direct sunlight
EV.6.2 EV.6.2.3	99 99	Insulation Add Clauses Heatshrink and similar insulation products must: a) Be marked with temperature rating and insulation voltage rating. A serial number or a norm printed on the wire is sufficient if this serial number or norm is clearly bound to the insulation characteristics for example by a data sheet. b) Have temperature rating more than or equal to 90°C All tractive system cable connections, splices, joins or terminations must be protected by a rigid enclosure (i.e. plug/socket housing, electrical enclosure or similar).
EV.6.2.4	99	Tractive System Wiring outside of electrical enclosures. In line with Clause EV.3.2.6, the energy meter HV DC measurement cable is exempted from the requirements of this clause. No additional covering beyond the existing white insulation sheath is required and should not be added.

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EV.6.4	99	CONNECTIONS
EV.6.4.3	99	Delete US words for EV.6.4.3 and replace with: Bolted electrical connections in the high current path of the Tractive System must include a positive locking feature to prevent unintentional loosening, with a rated operating temperature of at least 90 degrees C. <i>Lock washers or thread locking compounds (Loctite®) or adhesives are not acceptable. Bolts with nylon patches are allowed for blind connections into OEM components only where the OEM intended their use</i>
EV.6.4.5	100	For clarification add Additional Clause While the rules state specifically that anything in the high current path is a critical fastener, teams should also ensure that other fasteners inside the Accumulator Container do not loosen during vehicle operation. If they loosen or come undone, they must not pose a risk of a short circuit. A team does not need to positively lock a fastener if they can demonstrate that there is no likelihood of danger or a fire if the fastener fails (e.g. the fastener and the item it is securing are non-conductive). If there is a likelihood of a short circuit of fire if the fastener fails, it is classified as critical. The use of items such as glue, adhesive tape or Velcro are not considered acceptable as an alternative means of fastening.
EV6.5	100	Voltage Separation Delete US words for Clause EV.6.5.4 and Add
EV.6.5.4	100	GLV Systems other than the IRs EV.5.4, parts of the Precharge and Discharge Circuits EV.5.6, HV DC/DC converters, the BMS EV.7.3, the IMD EV.7.6, parts of the Tractive System Active Light EV.5.12, the Energy Meter EV.3.1 and cooling fans must not be inside the Accumulator Container.
EV.6.6	101	Overcurrent Protection Add Clause
EV.6.6.8	101	Overcurrent protection for the BMS voltage sense wires must be provided by a non-resettable fuses, with a voltage rating greater than the maximum segment voltage.
EV.7	102	Shutdown Systems Add revised words to include a PDOC circuit in the following clauses:
EV7.1.1	102	Add Clause D. Precharge/Discharge Overload Circuit (PDOC)
EV.7.1.3	102	The AMS, IMD, BSPD and PDOC parts of the Shutdown Circuit must be designed as Normally Open contacts
EV.7.1.4	102	The AMS, IMD, BSPD and PDOC must have completely independent circuits to Open the Shutdown Circuit. The respective circuits must be designed such that a failure cannot result in electrical power being fed back into the Shutdown Circuit.
EV.7.2.3	103	When the Shutdown Circuit is Opened by the AMS, IMD, BSPD or PDOC:

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EV.7.5	104	Accumulator Temperature
EV.7.5.5	105	Wording revised for clarification of intent. Delete US words and Add: For lithium based cells, while it is desirable to monitor the temperature of every cell, - The temperature of at least 20% of the cells must be monitored by the AMS. - The monitored cells must be equally distributed inside the Accumulator Container(s)
EV7.5.8	105	Add additional clause: An independent cell temperature monitoring device will be provided by the officials during accumulator inspection and must be installed, see IN4. The device must be placed on the negative terminal of one battery segment, such that the thermal strip will reasonably represent the same temperature as the <i>battery cells</i>. It must be in direct contact with the terminal or less than 30mm away from it on the busbar. The cell monitoring device will be a thermal sensitive sticker, maximum dimensions of 55mm long by 20mm wide by 4mm thick. Teams must provide an easy way for the thermal strip to be accessed during scrutineering and Parc Fermé, such as a hatch or transparent window in the accumulator housing. The maximum allowable window size must not exceed 30mm x 60mm and must be covered by a material equivalent to that defined in F.10.2, or a transparent material meeting F.1.18. This must not reduce the level of structural integrity or Ingress Protection.
EV.7.6	105	Insulation Monitoring Device Add the following words to the clause;
EV.7.6.2	105	In addition to the Bender ISOMETER® IR155-3203 or IR155-3204 as nominated in the Rules, the following alternative IMD's are approved for Formula SAE: - Bender ISOMETER® iso165C-1 – this IMD must not connect to the vehicle CAN bus - Bender ISOMETER® iso175 The following IMDs are Not Approved - Bender ISOMETER® iso165C - Sendyne/Sensata SIM100MOD - Sendyne/Sensata SIM100MLP - Orion BMS IMD.
EV.7.7	105	Brake System Plausibility Device – BSPD Add Two additional clauses:
EV.7.7.5	106	BSPD Circuit Test Teams must be able to prove the correct function of the BSPD circuit without spinning the vehicle's motors. This test must safely simulate power flow to the motors by injecting a test current directly into the main current sensor (through an auxiliary winding on this sensor for example), while the driver depresses the brake pedal. Teams should detail their test plan in their ESF and FMEA and will be required to demonstrate correct function of the BSPD during EV scrutineering

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EV.7.7.6	106	The status of the BSPD must be shown to the driver by a red indicator light in the cockpit that is easily visible even in bright sunlight. This indicator must light up, if the BSPD opens the tractive system shutdown circuit.
EV.8	107	CHARGER REQUIREMENTS
EV.8.1.3	107	Add clause defining the Charger Connector and Power Supply: Electrical power will be supplied for teams to recharge their vehicles via an AS3123 compatible 32 amp, 415 volt, three phase, 5 pin connector. The specific location(s) for the charging points will be advised in the Event Handbook. Teams that require a single phase supply for their chargers shall provide an appropriately tested and tagged breakout box or adapter cable that connects to the 32A 415V 5 pin connector that is provided.
EV.8.1.4	107	Add Clause defining Labelling Requirements The charger, accumulators and battery hand cart must be labelled clearly with the ESO's name and contact number. Letters shall be clearly legible, written in Roman Sans Serif numerals and at least 10mm high
EV.8.2.8	106	The AC power supply to the battery charger and other associated devices must include a residual current device (RCD) with over current protection (fuses or an appropriate circuit breaker) or residual current circuit breaker (RCBO). The RCD or RCBO device must act to disconnect both the active and neutral supplies. The trip sensitivity of the RCD must not exceed 30mA. Where possible 10mA is preferred.
EV.11.3	110	Lockout
EV.11.3.3	110	Add Clause For the lockout, teams should use an industry standard LOTO lock, not a generic padlock. There should be only one key and it should be in the custody of the ESO. The lock must be fitted at all times when the vehicle is not being operated

IN-TECHNICAL INSPECTION

US RULE	PAGE	CHANGES & CLARIFICATIONS
IN.2	112	INSPECTION CONDUCT Any other special requirements for Inspection at the Australasian event not listed in this Addendum will be detailed in the Event Handbook and/or on the website.
IN.4	113	ELECTRICAL TECHNICAL INSPECTION (EV) Clarification: At the FSAE-A event, the following procedure will apply to complement the requirements identified in IN-4 per the SAE International Rules. All inspection items per IN.4.1 must be brought to the inspection. Electric Vehicles Only Technical Inspection will be a sequential step process as follows:

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IN.4 (Cont'd.)	113	1. Verification of State of Charge (desirably should be 50% or less) and a visual inspection of all electrical systems and will involve internal inspection of battery packs and all HV system enclosures. Time will be scheduled Thursday and Friday for this inspection. 2. Mechanical Inspection as per the relevant parts of IN.8. 3. Tilt test may be completed after completing part 2. 4. Functional testing where teams will be required to demonstrate correct function of safety systems within the car and final electrical inspection. 5. Installation of the temperature indicating thermal strip and energy meter. Verification of Energy Meter Operation. 6. Teams will be required to complete the above steps before proceeding to brake test, dynamics events, or test pad 7. Teams are not to engage the HVD or power up their cars until their vehicle has advanced with sufficient level of sign-off and the team has been given specific approval to do so by the EV officials.
IN.4.8	114	Added Clause: BSPD Circuit Test BSPD Function and indicator light illumination will be checked in accordance with EV.8.7
IN.5.2	114	Egress Test Add clause:
IN.5.2.3	115	A list of the names of all drivers and times they achieved in the test must be provided by each team with the Technical Inspection List at on-site registration. The drivers who will be required to complete the test for verification at Technical Inspection will be identified at the event. This may or may not include all drivers.
IN.6	115	Driver Template Add Clause:
IN.6.3	115	To ensure adequate driver protection for varying driving positions, and to ensure a common approach to driver packaging, if the requirements of F5.6.3 to F5.6.5 are not met with the 95th percentile male template, 35 points will be deducted from the team's design event score and the car will not be allowed to compete in any dynamic events until modified to ensure compliance. The 915mm minimum dimension of the diagram in F5.6.5 must be maintained.
IN.8.1	115	Inspection Items Add the following items to the required list: • IC cars only with Electronic Throttle Control: ETC Review Form • The tested sample of the Standard IA if required to be tested due to the bulkhead configuration. • The bulkhead sample from the Impact Attenuator Test if not included with the Attenuator
IN.10.2.2	117	Sound Level Measurement (IC vehicles) – Clarification The US rules were modified in 2020 to adopt the Australasian approach to sound level measurement for special exhaust configurations but the following more specific test definition is provided for clarity:

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IN.10.2 (Cont'd.)	117	Height of the sound meter will be such that the vertical level will be determined by placement at an angle of 45° to a projected line of the centreline of the exhaust direction at the outlet. (This will not be at the height of the exhaust outlet except for exhausts exiting parallel to the ground.)
IN.11	118	RAIN TEST (EV ONLY) Add Clause
IN.11.4	118	Sealing Temporary sealing fixes, such as removable tape, will not be accepted as adequate vehicle design in order to pass the Rain Test.
IN.12	118	BRAKE TEST – IC and EV Add Note to Objective clause:
IN.12.1	118	Brake Light Check During the brake test IN.12.1 and IN.12.2 the officials will assess if the illumination is judged as satisfactory for clear observation in sunlight, by observation from the rear. This will be a subjective judgement.
IN.12.4	119	Add Clauses Tyre Pressure. During the Brake Test the tyre pressure must be set within the ranges designated below to ensure realistic pressures are used relative to the Endurance Event and the intent of the Brake Test. Dry Racing Tyres; 6 – 18 psi Wet Racing Tyres: 10 – 24 psi. The vehicle must pass the test using a pressure within this range. (Numbers in Green are preliminary and will be confirmed or updated)
IN.12.5	119	Tyre Type. The officials will try to run the Brake Test during “Dry” track conditions, however, if this is not possible and the track is declared “Damp” or “Wet”, the test must be run using the team’s Wet tyres.

S-STATIC EVENTS

S2 PRESENTATION EVENT		
US RULE	PAGE	CHANGES AND CLARIFICATIONS
S.2.2	121	Presentation Concept The Concept for the Presentation Event at FSAE-A 2021 will be available on the Formula SAE-A website and in the Event Handbook.
S.2.9 S.2.9.2	122	Scoring Add additional note re scoring method. A scoring Rubric will be released concurrently with the Presentation concept.
S.2.9.4	122	Add additional Clauses: At the Australasian Event a nominal timing of 10 minutes is allowed for the team’s initial presentation. Teams should try to operate within this guideline and Points will be deducted for significant variations to the guideline, Running under or over by 2 or more minutes will incur a Penalty of 2 Points. Running under or over by 4 or more minutes will incur a Penalty of 4 Points.

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S.2.9.5	122	Teams which do not meet the concept as required by S.2.2.3, may be penalised up to 50% of the score available.
S.2.9.6	122	The initial scoring rubric may be marked to a maximum score other than 75 points. The final scores will be adjusted so that the highest scoring team(s) will be awarded 75 points and all other scores will be adjusted in the same ratio, according to the formula Final Score = $SR(\text{your}) \times 75 / SR(\text{max})$ Where SR(your) is your team's initial score from the rubric and SR(max) is the score of the team(s) achieving the highest initial score from the rubric.

S.3 COST & MANUFACTURING EVENT

For the FSAE-A Event, the complete set of US Rules are replaced by the local event Rules which are detailed in Appendix PDA-2 of this Addendum.

S.4 DESIGN EVENT

The Design Score Sheet can be downloaded the SAE-A website at www.saea.com.au.

The details relating to the conduct of the Design Event will be communicated closer to the event date in the Event Handbook.

Some additional clarifications on Design Documentation and Formats may also be provided in the Event Handbook to be published later in the year.

S.4.2	126	Design Documents – Required Submission
S.4.2.1	126	Delete USA words and add revised wording: The Design Briefing, Vehicle Drawings, and Design Specification Sheet, must be submitted prior to the event via the online Formula SAE-A 2025 Document Submissions Google Form. https://forms.gle/mjrx1NDydSmAhByV8
S.4.3.1	126	Design Briefing Delete US Words and Add: The Design Briefing must follow guidelines defined in <i>Design Event Overview</i> document on the FSAE-A Website.
S.4.3	126	Design Documents Content and Format
S.4.3.4		Add extra clause: All of the Design Document files must be named as follows using the SAE-A assigned car number and the complete school name: • carnumber_schoolname_Design_Briefing.pdf <i>Example: 001_University of SAE_Design_Briefing.pdf</i>
S.4.11	127	Add Additional Clause Second Year Cars – Penalties for Insufficient Redesign Add the following clauses;

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S.4.11.1	127	The judges will deduct up to fifty (50) points from the final design score for cars which have not been substantially modified in a number of systems. If designs are new but similar, it is advisable to bring along photographic or other evidence of the level of change.
S.4.11.2	127	An additional thirty (30) points may be deducted from the design score if the team fails to show a balance of design between old and new, or if new parts have only been superficially altered, or the team members fail to show an overall understanding of the design rationale and detail of major components.

D-DYNAMIC EVENTS

DYNAMIC EVENTS The following general rules covering vehicle operation will apply at the FSAE-Australasia for all Dynamic Events and are supplementary to the published USA Formula SAE Rules.		
US RULE	PAGE	CHANGES & CLARIFICATIONS
D.1	128	General Dynamic
D.1.3	128	Add Clause D.1.3. Starting Time/Run Order In addition to Endurance, the officials will establish a Run Order by allocating and advising a Starting Time for each team in all Dynamic Event. Teams must be present in the starting queue by this allocated time or penalties may be applied, in the form of a time penalty. Any application of penalties will take into account the circumstances and adverse effects relating to the occurrence which may be beyond the control of the team(s). Deliberate running late by a team with the intent to gain an advantage will be more severely penalised. If penalties are applied, they will be within the following ranges- Acceleration: 0.2 - 0.4 seconds Skid Pad: 0.2 - 0.3 seconds Autocross: 5 - 8 seconds
D.3	129	Driving
D.3.2	129	Dynamic Area Limitations
D.3.2.4	129	Add Additional Clause: For all Dynamic Events at the Australasian Competition, a physical “Gate” will apply in the Staging Area. The running times for all Events as listed in the Handbook may be modified on the actual day subject to circumstances. The latest timing will be announced/displayed at the start of each day. Teams should present for each event as soon as ready and/or in accordance with any advised schedule. When ready to run they should enter the Staging Area and will be classified as “Inside the Gate”. No work other than adjustments not requiring tools may be performed on the vehicle inside the Gate. If any work is subsequently required, the vehicle must be taken outside the Gate.

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D.3.2.4 (Cont'd.)	129	Closing of the event will proceed via announcements that The Gate is closing 15 minutes before closure and at 5 minutes before closure. It is the responsibility of teams to ensure they are aware of these times. All vehicles inside the Gate and capable of running at the time of closure will be allowed to compete. Vehicles not inside the Gate in a ready to run condition at the Closure time will forfeit their right to compete or undertake any further runs in that event. Driving Under Power For clarification: Delete US words in D.3.3.1 and Add Vehicles may be driven under their own power only when inside the designated Dynamic Areas, (e.g. running in a dynamic event; on the practice track; during brake test) unless otherwise directed by an official. Driver Equipment Add Clause A “Percy” driver clearance check may be conducted in the staging area by the officials prior to drivers starting in any dynamic event or at any subsequent point if there are concerns regarding compliance with this safety requirements.
D.3.3	129	Add the following clause:
D.3.3.1	129	External Equipment and Work on Vehicles All vehicles must be capable of start, stop, restart and idle in all dynamic events, without external assistance, once the vehicle is on the starting line. This reinforces the requirement that any item essential to satisfactory vehicle operation are included in the cost and design reports for the event.
D.3.5	129	
D.3.5.3	129	Accordingly, for all dynamic events, from the time that the vehicle is deemed “ready to run” and has moved forward to the starting line under the starter’s control, it cannot be worked on and no auxiliary batteries or cooling fans are allowed, until the event is completed (including all heats required to be run consecutively or with some delay under officials’ direction).
D.3.11	130	If the vehicle subsequently cannot run it may be removed from the line and repaired but will be deemed to have run “out of order”. Additionally, to avoid disruption to the start line, ensure safe operation and not impair clear movement of other vehicles, the above requirements will also apply for vehicles entering the Staging Area queue inside the Gate for an event, unless specific clearance for any work or use of auxiliary equipment has been obtained from the officials controlling that event. Add additional clause: Dynamic Events – Remotely Changing Vehicle Specifications (Telemetry) In all Dynamic Events, once the vehicle enters the ‘hot’ or starting area under the official starter’s control and until leaving Parc Fermé (where this is applicable), teams are prohibited from transmitting any data to the vehicle that changes. Or advises the driver to change, the configurations/parameters of the vehicle. Changes as allowed under IN.14.2.3 are permitted, provided they can be safely performed without any tools on a moving vehicle. Radio communication is permitted between the driver and pit subject to ensuring safe installation of any such equipment.

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D.3.12	130	Contravention of this clause will result in zero score for the event concerned. Vehicle condition monitoring and communication with the driver is permitted.																																											
D.4	130	FLAGS The specific flags to be used at the Australasian event will be clarified at the event’s team and driver briefings. Green and red “lollypop” signals may also be used for signalling entry to the track. Add Clause																																											
D.4.3	130	In cases of excessive non-compliance with safety related flag directions, specific penalties will be applied to teams whose drivers are assessed as not complying with flag directions. These will be assessed by the officials but if a penalty is assessed as being required, the following penalties will generally be applied: - Failing to pull into the passing diversion under a Blue Flag direction – 5 Second penalty per flag point and/or up to a Black Flag. - Failing to slow down under a Yellow Flag direction – 10 seconds per flag point. - Failing to Stop under a Red Flag direction – 40 seconds per flag point and/or Black Flag. - Failing to obey a Black Flag – Will be assessed on number of non-compliances but may be held in the driver change area and could result in added time penalties and may include disqualification. Additional Post Endurance penalties may be applied post-event as covered under Clauses D.14.2 and D.14.3.																																											
D.6.2	131	Tyre Changes during Endurance																																											
D.6.2.3	131	The allowed tire changes and associated conditions are given in the following tables. In recognition of the severe degradation of Wet tyres running on a Dry track, a dispensation to allow a tyre change during Driver Change is added for the Australasian competition and the US Tables are replaced with the following condition and change tables. <table><tr><th rowspan="2">Existing Operating Condition</th><th rowspan="2">Currently Running On</th><th colspan="3">Operating Condition Changed to:</th></tr><tr><th>Dry</th><th>Damp</th><th>Wet</th></tr><tr><td>Dry</td><td>Dry Tyres</td><td>OK</td><td>A</td><td>B</td></tr><tr><td>Damp</td><td>Dry Tyres</td><td>OK</td><td>A</td><td>B</td></tr><tr><td>Damp</td><td>Wet Tyres</td><td>C</td><td>D</td><td>OK</td></tr><tr><td>Wet</td><td>Wet Tyres</td><td>C</td><td>D</td><td>OK</td></tr></table> <table><tr><th>Code</th><th>Requirement</th><th>Allowed at Driver Change?</th></tr><tr><td>A</td><td>May change from Dry to Wet</td><td>Yes</td></tr><tr><td>B</td><td>MUST change from Dry to Wet</td><td>Yes</td></tr><tr><td>C</td><td>May change from Wet to Dry</td><td>Yes</td></tr><tr><td>D</td><td>May change from Wet to Dry</td><td>No</td></tr></table>	Existing Operating Condition	Currently Running On	Operating Condition Changed to:			Dry	Damp	Wet	Dry	Dry Tyres	OK	A	B	Damp	Dry Tyres	OK	A	B	Damp	Wet Tyres	C	D	OK	Wet	Wet Tyres	C	D	OK	Code	Requirement	Allowed at Driver Change?	A	May change from Dry to Wet	Yes	B	MUST change from Dry to Wet	Yes	C	May change from Wet to Dry	Yes	D	May change from Wet to Dry	No
Existing Operating Condition	Currently Running On	Operating Condition Changed to:																																											
		Dry	Damp	Wet																																									
Dry	Dry Tyres	OK	A	B																																									
Damp	Dry Tyres	OK	A	B																																									
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C	May change from Wet to Dry	Yes																																											
D	May change from Wet to Dry	No																																											

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D.6.2.5	132	If the vehicle has a tyre puncture; Add additional clarification for Australasian event:
D.6.2.5.b	132	Delete USA words and Add: Teams that have incurred a puncture during the endurance event due to external factors (e.g. debris on track) may change the tyre within the driver change area, with no time penalty for the tyre change time. The wheel/tyre removed will be impounded and if, on inspection by the judges, it is subsequently assessed that the deflation/puncture was not caused by external factors, the vehicle will then be given a DNF for the event. Deflation or punctures caused by running off course or impacting barriers or other objects due to driver error will not be regarded as external factors.
D.9 ACCELERATION EVENT		
D.9.1.4	132	Delete US Clause and Add the following words. Cones will have their base position marked in all events, including Acceleration.
D.9.2	133	Acceleration Procedure
D.9.2.5	133	Add Clause The tyre pressure must be set at or below the maximum manufacturer recommended pressure.
D.11 AUTOCROSS EVENT		
D.11.1	135	Autocross Layout Clarification for local event: The track will generally be similar to the USA rules but teams will be advised of the final layout, the distance to be run for a heat and direction of travel prior to the event. Teams will have the opportunity to walk the track with the Clerk of Course on the Saturday of the event. Minimum track width will be 3.5m.
D.12 ENDURANCE EVENT		
D.12.1	137	Endurance General Information Add the following sentence to the US words:
D.12.1.5	137	The number of vehicles on the track simultaneously will be at the discretion of the Clerk of Course but generally will not exceed four.

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D.12.2	137	Endurance Layout Clarification for local event: The track will generally be similar to the USA rules but teams will be advised of the final layout and direction of travel on site, prior to the event. Teams will have the opportunity to walk the track with the Clerk of Course on the Saturday of the event. Minimum track width will be 3.5m.
D.12.12	140	Endurance Penalties Add additional clause:
D.12.12.7	140	Penalties will not be assessed for accident avoidance or other reason deemed sufficient by the track officials. Adjustments to elapsed time may be made for cases where teams may be halted or disrupted by another team, or by track officials; such adjustments will be entirely at the discretion of the judges/track officials.
D.12.13.3	138	Lap Score for Endurance Added Note: As the full number of laps for Endurance at the FSAE-A Event may not be exactly 25 laps, in order to achieve the maximum score of 275 points for the fastest team(s), completing all laps, an adjustment will be made for the completed laps score in accordance with the following formula, where • 'L' is the full number of laps at FSAE-A and • Laps(your) is the number of completed laps for eligible vehicles $\text{Laps Score (your)} = \text{Laps(your)} \times \{25 / L\}.$ The resultant score will be rounded to one decimal place. This number is added to the Time Score to achieve the overall Endurance Score.

D.13 Efficiency Event

For the FSAE-A Event, the complete set of US Rules for Efficiency are replaced by the local event Rules which are detailed in Appendix PDA-3 of this Addendum.

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CHANGE LOG

S.3.2.2; S.3.3.3;	Minor word revisions
S.3.3.4.1;	Examples deleted. Will be covered in training seminar
S3.12.2.a; S.3.13.6; S.3.17.1;	Minor word changes and eliminate anomalies.
EV.5.10; EV.5.11	Deleted US rules for Tractive System Status Indicator and Ready to Move Light
EV.5.12	Reinstated 2024 Rules for Tractive System Active Lamp (TSAL)
EV.6.5.4	Revise words to reflect reversion to TSAL versus Ready to Move light.
F.10.3	Accumulator Cells and Segments. Reference to Guideline added to the website to assist with SES input.
F.8.2.3; F.8.5	Clarification of requirements for use of a Quick Release for the AIP to Bulkhead and note on preferred method of IA attachment.

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APPENDIX PDA - 1

Action Deadlines for 2025 Formula SAE Australasia

All submissions must be uploaded via the online Formula SAE-A 2025 Document Submissions Google Form by 5:00 PM (Melbourne local time) on the defined date. Teams should check and allow for time zone and Summer/Standard/Winter-time differences.

The US Rules for late receipt apply, except where otherwise noted earlier in this Addendum. Forms and templates and details of their required format can be downloaded from the SAE-A website at <https://www.saea.com.au/rules-documents-templates>.

All electronic submissions are to be uploaded by the Team Leader to the online [Formula SAE-A 2025 Document Submissions Google Form](#) using a University or official team email address. The Team Leader email addresses must be unique for Universities with multiple entries. An email acknowledging receipt will be provided by the Google Forms site. Use the included edit link to submit subsequent documents.

Dates for Autonomous Vehicle teams and Submissions are common with other Dates as shown for the applicable powertrain type. Where unique AV submissions/documentation are required they are shown as a separate line item. The dates shown in Green are provisional only and may be corrected from this initial issue.

Submissions must adhere to standard file naming and file format listed in Table DR-1.

Other than what is specified for on-site registration, no hard copy submissions will be required.

Date	Milestone/Deadline	Submission Method	Vehicle Type	Event Type
14 Apr	Registration Open for all teams. Registration and payment may be submitted.	Electronic	EV & IC	Dynamic & Static
2 May	Electronic Throttle Control (ETC) Deadline for Notice of Intent submission	Electronic	IC	Dynamic
20 Jun	Final date for submission of Application to Compete with 2 nd Year Car	Electronic	EV & IC	Dynamic and Static
8 Aug	All Teams: (i) Team Registrations Close (ii) Entry Fees deadline	-	EV & IC	Dynamic & Static
22 Aug	Electrical systems officer and electrical systems advisor forms deadline	Electronic	EV	Dynamic
22 Aug	ESF deadline	Electronic	EV	Dynamic & Static
22 Aug	Autonomous Systems Form	Electronic	EV (AV)	Dynamic & Static
22 Aug	Rationale/Evidence for non-fitment of PDOC	Electronic	EV	Dynamic & Static
22 Aug	Electronic Throttle Control Systems Form	Electronic	IC	Dynamic
22 Aug	Autonomous Vehicle Operation Diagram	Electronic	EV & IC (AV)	Dynamic & Static
22 Aug	Autonomous Vehicle Wiring Diagram	Electronic	EV & IC (AV)	Dynamic & Static
22 Aug	Autonomous Vehicle Schematic Diagrams	Electronic	EV & IC (AV)	Dynamic & Static
12 Sep	Structural Equivalency Spreadsheet deadline including Integral IAD report	Electronic	EV & IC	Dynamic & Static

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12 Sep	Autonomous Vehicle Mechanical Parts	Electronic	EV & IC (AV)	Dynamic & Static
12 Sep	Autonomous Hydraulic/Pneumatic Systems	Electronic	EV & IC (AV)	Dynamic & Static
19 Sep	AV teams notice of entry in Static Event(s)	Electronic Word Doc.	EV & IC (AV)	Static
3 Oct	Cost Report and Support Materials deadline	Electronic	EV & IC	Static
3 Oct	Design Briefing, Specifications Sheet and Vehicle Drawings deadline	Electronic	EV & IC	Static
3 Oct	2 nd Year car documentation, photograph and details of modification with FA Statement.	Electronic	EV & IC	Static
10 Oct	Motorsport Australia license application submission deadline	Electronic	EV & IC	Dynamic
10 Oct	Declaration of planned hazardous materials and MSDS deadline.	Electronic	EV & IC	Dynamic
10 Oct	Summary of Risk Assessment, Hazard Reduction and Safe Working practices.	Electronic Word or PDF	EV & IC	-
17 Oct	Cost Scenario Task Advised to Teams	Email and Website	EV & IC	Static
07 Nov	Final team member list deadline	Electronic	EV & IC	Dynamic & Static
07 Nov	Individual team member registration and fees deadline (All teams).	-	EV & IC	Dynamic & Static
07 Nov	Autonomous Systems Design Report	Electronic	AV	Static
01 Dec	Tech Inspection Checklist including notice of ETC approval; Egress Times List.	Electronic	EV & IC	Dynamic
01 Dec	Electrical Inspection Checklist deadline	Electronic	EV	Dynamic
01 Dec	Autonomous Vehicle Inspection Checklist	Electronic	EV & IC (AV)	Dynamic
11 Dec	Bump-In and Team Registration is open from 3pm to 5pm	On Site		
Tues 9 Dec	Cost Amendment Report Final Submission Deadline	Electronic	EV & IC	Static
11 Dec	Declaration of final hazardous materials and MSDS deadline; copy of Tech Inspection checklists; copy of Egress Times List.	Hand Deliver at Site Registration	EV & IC	Dynamic
Wed 9 Dec	Nomination of Focus Cost System(s) to team	Advised at Registration	EV & IC	Static
11 Dec	Signed confirmation of compliance with safety standards and safe working practices for EV's.	Hand Deliver at Site Registration	EV	-
11-14 Dec	Formula SAE-Australasia Competition Team Registration is open from 8am to 5pm	On Site		

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Rule and Cost enquiries are to be submitted online via the process outlined on the SAE-A website.

<https://www.saea.com.au/submit-a-rules-enquiry>

General enquiries may be submitted via email to formulasae@sae-a.com.au

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Cost & Manufacturing Event Rules 2025 V1.0

S.3 COST AND MANUFACTURING EVENT

S.3.1 Cost Event Objective

S.3.1.1 The Cost and Manufacturing Event is intended to assess the abilities of each team in relation to manufacturing engineering knowledge and capability, project management with respect to cost-based parameters, and decision trade-offs made between the vehicle manufacturing cost, profitability for the business enterprise per rule GR.1.4.1, and the dynamic performance of the vehicle. Making trade-off decisions between content and cost, based on the performance of each part and assembly, and accounting for each part and process to meet a budget is a critical part of Project Management.

S.3.2 Cost Event Supplementary Information

S.3.2.1 A Cost and Manufacturing Event guide will be published to the SAE-A website containing an overview of costing process and how to fill in the cost report part and assembly data tables.

S.3.2.2 The Cost Catalogues (S.3.8) will be available from the SAE-A website only. Teams should ensure they use the latest version of the catalogues when developing their cost reports.

The catalogues will not be changed for any Add Items Requests, or other requests, from a period 2 calendar weeks in advance to the cost report submission deadline.

S.3.2.3 Submission links for the Cost Report (S.3.4), Cost Amendment Report (S.3.7), and Cost Add Item Requests (S.3.10) will be made available to each registered team by the SAE-A.

S.3.2.4 A Cost and Manufacturing Event Description will be published in the FSAEA Event Handbook, containing an overview of how the event is expected to run, what the specific requirements are to be met during the event for the team being scored.

S.3.2.5 A Cost and Manufacturing Event Timetable will be published in the FSAEA Event Handbook, and teams must present at the cost event during their allocated time slot.

Allocated time slot changes will only be authorised in exceptional circumstances outside the control of the team, and subject to the Officials' discretion.

S.3.3 Cost Event Areas

S.3.3.1 Cost Report Preparation and submission of a report (the "Cost Report")

S.3.3.2 Event Day Discussion at the Competition with the Cost Judges around the team's vehicle.

S.3.3.3 Cost Scenario where teams will respond to a challenge related to cost or manufacturing [engineering or engineering business integration](#).

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Cost & Manufacturing Event Rules 2025 V1.0

S.3.4 Cost Report

S.3.4.1 The Cost Report must:

- a. List and cost every part on the vehicle using the standardized Cost Catalogue of Materials, Processes, Fasteners, Tooling, and Multipliers (refer S.3.8).
- b. Be based on the actual manufacturing processes used to develop the prototype, with deviations for bulk manufacturing process changes allowed, provided the change is supported with supplementary evidence to support equivalency of the change.

Examples will be made available during the competition year, showing what is expected to achieve deviations, and which processes are expected to show evidence as part of this deviation process.

- c. Include Tooling Cost (welding jigs, moulds, patterns, and dies) for processes requiring it.
- d. Exclude R & D and capital expenditures (plant, machinery, hand tools and power tools).
- e. Include supporting documentation to allow officials to verify part costing.
 - i. For IC vehicles, teams must present at minimum the technical datasheet for the Engine, ECU, and injector/s.
 - ii. For EV vehicles, teams must present at minimum the technical datasheet for the Accumulator cells, the Battery Management System, the Tractive Motor/s and Controller/s, the main vehicle control module (ECU), and the low voltage battery pack/cells. Additional datasheets may be required for sensors and additional modules if costing of those components can be affected by features of those components.
- f. Include a summary with regards to the project cost management, including:
 - i. A brief summary of decisions made with regards to vehicular performance vs manufacturing costs;
 - ii. A summary of bulk manufacturing methods used, and how each is applied throughout the cost report.
 - iii. A summary of the part numbering convention used in the cost report.

S.3.4.2 The Cost Report must be submitted in a single PDF Document containing all relevant documentation. The PDF Document is considered the “Cost Report”, and must be submitted to the online submission form. The PDF document should include the cost report summary, the full vehicle Bill Of Materials, the cost section breakdown, all part and assembly data tables, all drawings and supporting documentation, and all datasheets for critical components only.

- a. Cost section breakdowns should be found on the leading page for each system, which in themselves should be clearly separated.
- b. Drawings and supporting documentation are expected to be in close proximity to the part and assembly data tables which refer to the documentation, and
- c. datasheets are expected to be located in an addendum at the end of the cost report.

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Cost & Manufacturing Event Rules 2025 V1.0

Supporting Excel Documentation (which may be submitted as multiple files) **must** also be submitted via the online Formula SAE-A Document Submissions Google Form provided by SAE-A, as per S.3.2.

It is highly recommended that these files are either:

- a. Submitted as individual excel sheets per part/subassembly/assembly;
- b. Submitted with a macro enabled worksheet with a cover BOM with links to the correct tab and a return to home button; or
- c. Submitted as a database with a searchable display page.

S.3.4.3 The Cost Report PDF document must be limited to <9.0GB in size, due to document submission limitations. If a cost report is, by content inclusion, larger than this size, alternative methods for data transfer must be arranged before the submission deadline.

S.3.5 Bill of Materials – BOM.

S.3.5.1 The BOM is a list for every vehicle part, showing the relationships between the items.

- a. The overall vehicle is broken down into separate Systems.
- b. Systems are made up of Assemblies.
- c. Assemblies are made up of Parts and/or Subassemblies.
- d. Subassemblies are made up of Parts and/or other Subassemblies.
- e. Parts consist of Materials and/or Fasteners, and Processes.
- f. Tooling is associated with each Process that requires production tooling.

S.3.5.2 The **entire vehicle** BOM is to be integrated into the cost report submission, and **must** be included in its entirety at the beginning of the cost report.

S.3.5.3 An assembly tree is to be included at the beginning of each system within the cost report **where systems contain more than 2 levels of parts within the system (i.e. if the system contains any subassemblies)**. These assembly trees showing the logical progression from each part through subassemblies to the completed system. **An assembly tree is not required in any system where only parts and high level assemblies are present.**

S.3.5.4 All parts submitted to the BOM must have matching part numbering to the respective part and assembly data tables and to their respective drawings.

Part Numbering must follow a standardized part naming convention standard. This numbering should produce a code which will be able to positively identify the part with regards to all other parts in the vehicle cost report with a string of characters, ranging in length from 6 to 12 characters, within the part numbering.

It is strongly recommended that the part numbering follow the following format **or the FSAE-I part numbering format**. The characters shown in yellow in the below example are the reference ID which is used by the cost judges to identify errors in your cost accuracy sheets, and no extra time to locate parts will be provided during the cost event based on the use of a complex part numbering convention.

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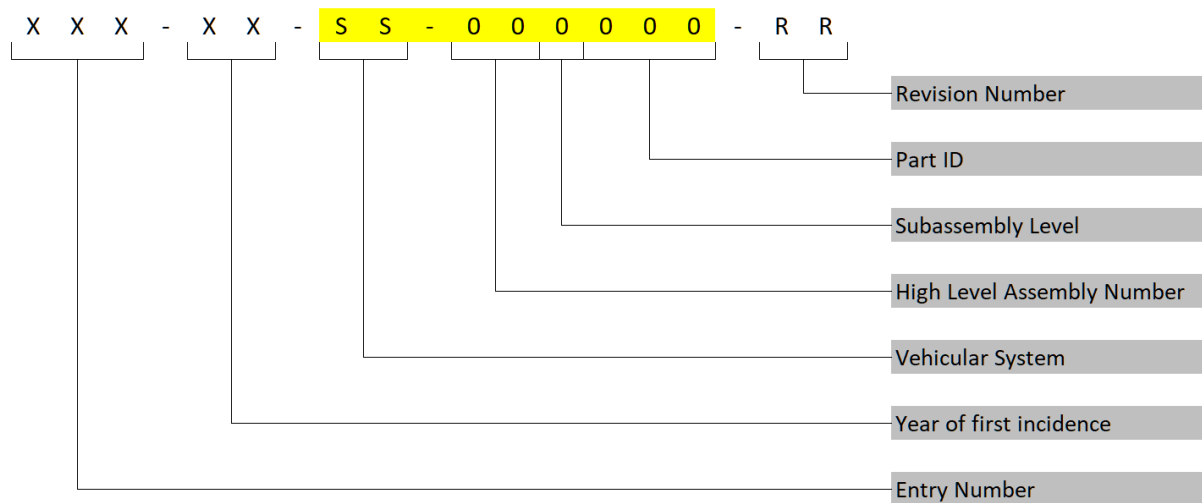


Image S3-1: Recommended part numbering convention for the FSAE-A competition

S.3.5.5 The cost and manufacturing event will not penalize teams within the cost and manufacturing event scoring based on the number of parts which are carried over, i.e. are of a year of first incidence other than the competition year at which the vehicle is presented.

S.3.6 Late Submission

S.3.6.1 Penalties for late submission shall be imposed as follows:

- For each business day past submission, a penalty of -5 points will be applied to the team's final cost report score, up to 10 business days past the submission deadline.
- Any submission not made before close of business on the 10th business day following the original submission deadline will not be eligible for any Cost Report score but may still compete in the Cost Task and Event Day Cost Challenge.

S.3.6.2 An incomplete submission will be defined as any submission which is missing either 1 or more full section of part and assembly data tables, or with over 20% of part and assembly data tables missing. This does not include Autonomous specific parts or assemblies unless the vehicle is registered as an "Autonomous" or "Dual Purpose" vehicle. Incomplete submissions will be treated as equal to a failure to submit, and penalties will accrue accordingly per S.3.6.1.

S.3.7 Cost Amendment Report

S.3.7.1 A supplementary Cost Amendment Report (formerly known as the cost addendum) may be required to be submitted at the event competition if any design or technical changes occur on the prototype between the original cost report submission and the cost event. The intent of this report is to document changes which occur during the prototyping, testing, and development phases of the vehicle project, and the design evolution thereof, and is not intended to be used as a method of completing an incomplete or inaccurate submission of the original cost report.

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S.3.7.2 The Cost Amendment Report is to be submitted in PDF format electronically. This submission may be made at any point following the original cost report submission, however it must be made no later than 11:59pm on the Tuesday immediately preceding the FSAE-A competition. Teams are advised that they are only permitted a single Cost Amendment Report Submission, and should plan any submissions accordingly. The cost amendment report will not be accepted past this deadline to prevent tailoring of reports to the target systems for the cost judge marking.

S.3.7.3 A specific format for the Cost Amendment Report will be published on the Formula SAE-A website.

S.3.7.4 Cost Amendment Reports apply only to the competition at which they are submitted.

S.3.7.5 Changes to the vehicle costing resulting from changes made in the Cost Amendment Reports will incur additional cost:

a. Added items will be costed at 105% of the catalogue cost: + $(1.05 \times \text{Cost})$

b. Removed items will be credited 95% of the catalogue cost: - $(0.95 \times \text{Cost})$

This cost delta change must be applied at the BoX (Material/Process/Fastener/Tooling) summed total price within the cost data tables, not at the system level. The cost delta must be summarised in a cover sheet, indicating part data (number and description), original BOM data (QTY, BoX prices, and total part cost), and the revised BOM data (QTY, BoX prices, and total part cost).

S3.7.6 The amendment report costed values and credited values have been significantly lowered compared to other competitions. This is to encourage teams to improve the accuracy of the cost report (with amendment report) to the presented vehicle, with modifications which were deemed technically necessary between the time of cost report submission and the competition start date.

Teams are warned that a poor-quality initial cost report with an excessively large amendment report will be subject to additional penalty if they are found to have used this rule outside of this intent, which will be scaled between the loss of event points to overall rejection of the cost amendment report for violation of intent.

S.3.8 Cost Catalogues

S.3.8.1 All costs in the Cost Report must come from the standardized Cost Catalogue. The Cost Catalogue for the FSAEA competition is found on the Formula SAE-A website, per S.3.2.2.

S.3.8.2 If a team wishes to use any Parts, Processes or Materials not included in the catalogue, an Add Item Request must be submitted. See S.3.10

S.3.8.3 Additionally, standardised stock sizes are included within the cost catalogues, and will be applicable to any purchased stock of a raw material. All raw materials of which there is a standardized stock size been used must use one of these stock sizes, with any none-standard stock size been extruded as though a custom extrusion, or otherwise formed.

Additional stock sizes may be added to the catalogue, but require an approved Cost Add Item Request to be used.

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S.3.9 Make versus Buy

S.3.9.1 All part may be classified as Made or Bought.

S.3.9.2 Some specific parts must be cost as made irrespective of if they are purchased on the prototype. Refer to the SAE-A Website for the latest Cost Catalogue for reference if a part is available for use.

S.3.9.3 If a team genuinely Makes a part listed on the Cost Catalogues as a Bought part, [and are the Intellectual Property \(IP\) owner for this part](#), they may alternatively cost it as a Made part

S.3.9.4 Any part which is normally purchased that is optionally shown as a Made part must have supporting documentation submitted to prove team manufacture.

S.3.9.5 Teams costing Bought parts as Made parts will be penalized. Penalties will be imposed based on the cost of the Bought part, at a cost of 125% of the catalogue price added to the reported price in the cost report.

S.3.10 Add Item Request

S.3.10.1 For the Formula SAE-A event, any Add Item Requests must be submitted via the online FSAEA Cost Add Item Request (CAIR) Google Form, for review and processing by the local Formula SAE-A Cost Committee.

S.3.10.2 A link will be provided to teams to access the Cost Add Item Request Form (CAIR), per S.3.2.2.

S.3.10.3 After review, the item may be added to the Cost Catalogue with an appropriate cost. It will then be available to all teams.

S.3.11 Public Cost Reports

S.3.11.1 The competition organizers may publish all or part of the submitted Cost Reports.

S.3.11.2 Cost Reports for a given competition season will not be published before the end of the [following](#) calendar year. Support materials, such as technical drawings [or team generated datasheets](#), will not be released.

S.3.12 Cost Report Penalties Process

S.3.12.1 For the Australasian competition, a two-step cost report penalty process will be followed. The process will be as follows:

a. The raw cost of the vehicle will be used as a base value in which 2 multipliers for accuracy and documentation submission quality will be applied to determine the penalised cost value of the vehicle.

$$P_{(\text{your})} = P_{(\text{reported})} * D * A$$

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where

- i. “P(your)” is your penalised vehicle cost.
- ii. “P(reported)” is the vehicle cost as reported in the cost report submitted and after processing of any changes made in the Cost Addendum and the targeted component penalty process, if applicable.
- iii. “D” is a multiplier used to adjust the cost based on documentation inadequacies and assessing the readiness for manufacturing of your vehicle. This multiplier represents the additional costs produced downstream due to the lack of completeness of the vehicles documentation. The multiplier is calculated as the multiple of the 4 sub-multipliers D1 through D4, as below.

$$D = D_1 * D_2 * D_3 * D_4$$

- iv. “A” is a multiplier used to adjust the cost based on the accuracy of the costing calculations presented in the cost report, using a combination of subsections in the vehicles, as described in S.3.17, to determine an estimated accuracy level for the entire cost report submitted. This multiplier represents the additional costs incurred into the vehicle mass production build due to the inaccuracies present in the cost report data, nominally through process and capital planning and execution.

b. For targeted components, where a material is under-quoted at a value greater than or equal to \$500, or a process is under-quoted at a value greater than or equal to \$50, the cost difference will be added to the reported cost, P(reported), and a ten (10) point deduction in the team’s specific accuracy multiplier will be incurred. Targeted components should be assumed to be based on the list of parts requiring datasheets, as per S.3.4.1e, composite monocoques, wheel rims, and tyres.

c. Where missing parts and/or assemblies are identified in the cost report, the following penalties will be added to cost report score post calculation of the cost report section score (refer S3.3.1), using the P(your) value shown above. This deduction will be referred to as the second step penalty.

- i. Missing Hardware – 0.5 points
- ii. Missing Part -2 points
- iii. Missing Assembly -5 points

S.3.12.2 a. For the purposes of the **D3 and D4 Documentation** sub-multiplier, the FSAEA competition will only require one to two (1-2) system/s to have fully prepared drawings for the grading of this rubric. These systems will be independent of the Accuracy Assessment systems, per rule S.3.17.

b. For the **2025 FSAEA** competition, the nominated system shall be **ST (Steering)**.

c. Any system which is not part of the nominated drawing system must still provide enough information visually for the cost judges to identify, visualise, and verify the manufacturing processes proposed for the part/assembly. It is highly recommended to meet this clause that

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teams provide either basic drawings or increase the general quality of their images/renders to allow for dimensions and detail views to be present in these images.

S.3.12.3 After calculation of the adjusted vehicle costs using the formula in S.3.12.1, any vehicle price which exceeds a value of:

- a. four times (4x) the minimum adjusted vehicle cost for IC vehicles, or
- b. three and one-half times (3.5x) the minimum adjusted vehicle cost for EV or Dual Purpose vehicles;

will receive 0 points for the cost report element of the cost event.

Based on prior competition years, this cutoff should be assumed to be approximately:

- a. \$32,500 +/- \$2,500 cost report price for IC vehicles; or
- b. \$100,000 +/- \$10,000 cost report price for EV/Dual purpose vehicles.

S.3.13 Event Day and Discussion

S.3.13.1 The team must present their vehicle at the designated time.

S.3.13.2 Teams must bring a computer with USB Type A Port capable of running the Cost Report, and any Cost Amendment Report, to the judging. The judges will provide a USB stick loaded with the team's electronic submissions relevant to the cost event.

A computer or other media may be used for presentation of other materials at the event as well as part or assembly data tables, charts, and drawings. Refer to Event Handbook for final details of the FSAE-A Cost Event.

S.3.13.3 The event day discussion will be broken down into 3 specific sections. These will be:

- a. A discussion on the cost report as submitted, with an opportunity for teams to challenge potential errors identified in the cost report by the cost judges, and which will ultimately be used to determine the "A" multiplier, as per S3.12.1.
- b. A Real-time Costing Challenge, where the team will be asked to cost a small assembly and identify what materials, processes, fasteners, and tooling would be required to manufacture the components and assemble, unaided by the cost report submission.

This challenge is to assess the teams specific manufacturing engineering knowledge, specifically relating to process knowledge and reverse engineering capabilities.

- c. The Cost Scenario presentation (S.3.14).

S.3.13.4 The real-time costing challenge and the cost scenario will have event score points attributed to both sections independently. The weighting of these sections will be published in the FSAEA Event Handbook.

S.3.13.5 The Cost Judges will:

- a. Review whether the Cost Report accurately reflects the vehicle as presented, treated as the prototype for a production vehicle.
- b. Review the manufacturing feasibility of the vehicle.

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- c. Assess supporting documentation based on its quality, accuracy, and thoroughness.
- d. Apply penalties for missing or incorrect information in the Cost Report compared to the vehicle presented at inspection.

S.3.13.6 Prior to the cost event all teams are expected to send a student representative to meet for a Cost Briefing, where all details of the event and any updates to the event will be mentioned. The student representative must be either the cost lead, or the student team leader. [Faculty advisers are optional attendees but are encouraged to attend this briefing to ensure they understand the operational requirements of the cost event.](#)

S.3.14 Cost Audit

S.3.14.1 Teams may be selected for additional review to verify all processes and materials on their vehicle are in the Cost Report. This may be completed as part of a specific request for assistance from a cost judge, or as part of the cost judge moderation process.

S.3.14.2 Adjustments from the Cost Audit will be included in the final scores.

S.3.15 Cost Scenario

The Cost Scenario will be provided prior to the competition on the FSAE-A Online website – refer to Appendix PDA-1. The Scenario will challenge teams to generate a proposal based on cost and/or manufacturing, and in relation to technical characteristics and business strategy and integration. The team's proposal is expected to be well reasoned, from the initial challenge statement through to the ultimate proposal recommendation, and must include all data used at each step of the process used to generate the recommendation.

S.3.16 Cost Event Scoring

S.3.16.1 The cost report score will be issued as a score out of 80. The cost report score, prior to the omitted item penalisation strategy, will be issued based on the vehicles cost relative to other vehicles in the competition, using the below formula.

$$\text{Cost report score} = 80 * \frac{(P_{\text{max}} - P_{\text{your}})}{(P_{\text{max}} - P_{\text{min}})}$$

Where,

- i. P_{max} is the highest vehicle cost post penalization in the current year competition.
- ii. P_{min} is the lowest vehicle cost post penalization in the current year competition.
- iii. P_{your} is your vehicle cost as per as per S.3.12.1

S.3.16.2 The real-time costing challenge score will be issued as a score out of 5, based on the discretion of the cost judging team on assessment that the selected assembly could be produced using the materials, processes, fasteners, and fixtures given during this discussion.

S.3.16.3 The cost scenario score will be issued as a score out of 15, based on the discretion of the cost judging team against a marking rubric, which will be advised in the Event Handbook and on the Formula SAE-A website.

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S.3.17 Cost Report Assessment Strategy

S.3.17.1 At the Australasian Event, teams are required to cost and submit a complete cost report per rule S.3.4.2, however judging at the on-site event will review only a specific subset of systems in detail. The chosen systems will be formally advised at the cost briefing before the cost event, and the team's specific accuracy sheet will be distributed to the team electronically at a specific time prior to the teams specific cost event allocated timeslot. The system subset will consist of 1 of 4 system groupings, as follows.

- a. DR (Engine/Tractive Path and Drivetrain) and EL (Electrical).
- b. CH (Chassis) and MS (Miscellaneous, Fit, and Finish).
- c. SU (Suspension), ST (Steering), and BR (Brakes).
- d. AD (Aerodynamics) and WT(Wheels and Tyres).

Judges will assess no less than 100 parts and their associated assemblies within these systems to ensure an adequate section of the cost report is reviewed for the A multiplier determination. If any grouping of systems found above does not contain enough parts to complete this assessment, the judges will select enough parts and associated assemblies from any of the remaining systems to complete this requirement, excluding parts located within the AV (Autonomous Vehicle and Control) System.

Additionally, vehicles registered as an "Autonomous" or "Dual Purpose" vehicle may have the AV (Autonomous Vehicle and Control) System section audited, at the discretion of the FSAEA competition organizing body and the FSAEA Cost Committee. Any findings from these sections will not be applied in the EV or IC class of a "Dual Purpose" vehicle but may be used in any cost related activity relating to the Autonomous Class.

S.3.17.2 Additional high value components may be audited outside of the system groupings shown above, at the discretion of the current years cost judging team.

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Efficiency Event Rules 2025 V1.0

D.13 EFFICIENCY EVENT

D.13.1 Efficiency Event General Information

- D.13.1.1** The Efficiency event evaluates the energy used to complete the Endurance event
- D.13.1.2** The Efficiency is derived from the chemical stored thermal energy of the fuel consumed for IC vehicles, or stored electrical energy used for EV vehicles, and the overall penalty adjusted time taken to complete the endurance course.
- D.13.1.3** No adjustment to distance or fuel/energy will be made.

D.13.2 Efficiency Event Procedure

D.13.2.1 For IC vehicles:

- a. The fuel tank must be filled to the fuel level line (IC.5.4.5)
 - b. During fuelling, once filled to the scribe line, no shaking or tilting of the tank, fuel system, or of the entire vehicle is allowed.
 - c. The vehicle will then compete in the Endurance event, refer to D.12.5
 - d. Vehicles must **shut down the engine** immediately upon leaving the course, and be pushed by the team to the 'Parc Fermé' station.
 - e. The fuel tank must be filled to the fuel level line (IC.5.4.5) to measure fuel used. IC.5.5.1.
 - f. If the fuel level changes after refuelling:
 - Additional fuel will be added to return the fuel tank level to the fuel level line.
 - Twice this amount will be added to the previously measured fuel consumption.
 - g. The officials may selectively shake/vibrate/tilt a vehicle at refuelling following endurance, or in the 'Parc Fermé'.
- In the event of any change in level, Clause D13.2.1.f penalties will be applied.

D.13.2.2 For Electric vehicles:

- a. The vehicle may be at any state of charge as presented at the entry to the endurance event.
- b. The vehicle must have a correctly installed energy meter present in the tractive path of the powertrain. Correct function of the energy meter should be verified before entry to the Endurance event.
Correct function of the energy meter will be verified during Technical Inspection.
- c. The vehicle will then compete in the Endurance event, refer to D.12.5.
- d. Vehicles must power down after immediately upon leaving the course and be pushed to the 'Parc Fermé' station.
- e. Energy meters must be removed from the vehicle as soon as possible in Parc Fermé, and passed to race officials for processing.
- f. Energy meter data will be downloaded to determine energy used and check for power violations, per EV.4.4. Penalties will be applied per EV.4.5.

D.13.3 Efficiency Eligibility

- D.13.3.1** Maximum Time.
Vehicles who finished endurance under the endurance event Tmax time shall be eligible for efficiency under time criteria.
- D.13.3.2** Maximum Fuel/Energy Used.
- a. IC Vehicles whose corrected average fuel consumption per lap exceeds 26 litre/100 km will **be ineligible for the efficiency event and** receive zero points.

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b. EV Vehicles whose corrected average energy consumption per lap exceeds an energy equivalent of 0.4 kWh/km will be ineligible for the efficiency event and will receive zero points.

D.13.3.3 Completion of Endurance.

a. Vehicles must complete the Endurance event and cross the finish line on their final lap to be eligible for efficiency under the completion criteria.

b. All other vehicles will be ineligible for the efficiency event and receive zero points.

D.13.3.4 Vehicle Condition.

If damage or a potential environmental or safety hazard (such as fuel tank leakage) exists, the fuel tank will not be refilled, and the team will be ineligible for the efficiency event and receive zero points.

D.13.4 Efficiency Scoring

D.13.4.1 Conversion Factors Each fuel or energy used is converted using the factors:

- | | |
|-----------------------------------|--------------|
| a. Petrol, 98RON | 32,317 kJ/L |
| b. Race Blend Petrol/Ethanol, E85 | 22,900 kJ/L |
| c. Electric | 3,600 kJ/kWh |

D.13.4.2 EV vehicles will be scored based on their net energy consumption over the endurance event, and full credit will be given to any EV vehicle for energy recovered through regenerative braking during the endurance event.

D.13.4.3 Scoring Term Definitions:

- | | |
|-------------|--|
| T_{min} | The corrected Endurance time of the team which maintained the highest average speed in the Endurance event and which is eligible for Efficiency. |
| T_{your} | The corrected Endurance time of the team being scored in the Endurance event. |
| E_{min} | The calculated energy used by the team for the entire Endurance event with the smallest energy consumption per lap, which is also eligible for Efficiency. |
| E_{your} | The calculated energy used by the team being scored for the entire Endurance event. |
| EF_{min} | The efficiency factor of the team with the lowest (worst) efficiency factor. |
| EF_{max} | The efficiency factor of the team with the highest (best) efficiency factor. |
| EF_{your} | The efficiency factor of the team been scored. |

D.13.4.4 Scoring Formula:

a. The Efficiency Factor (EF) for each eligible team will be calculated using the following formula.

$$i. \quad EF_{your} = \frac{T_{min}}{T_{your}} * \frac{E_{min}}{E_{your}}$$

b. The Efficiency Score will be calculated using the following formula.

$$ii. \quad \text{Efficiency Score} = 100 * \frac{EF_{your} - EF_{min}}{EF_{max} - EF_{min}}$$